Signal intelligence architecture: the driving-stack comparison

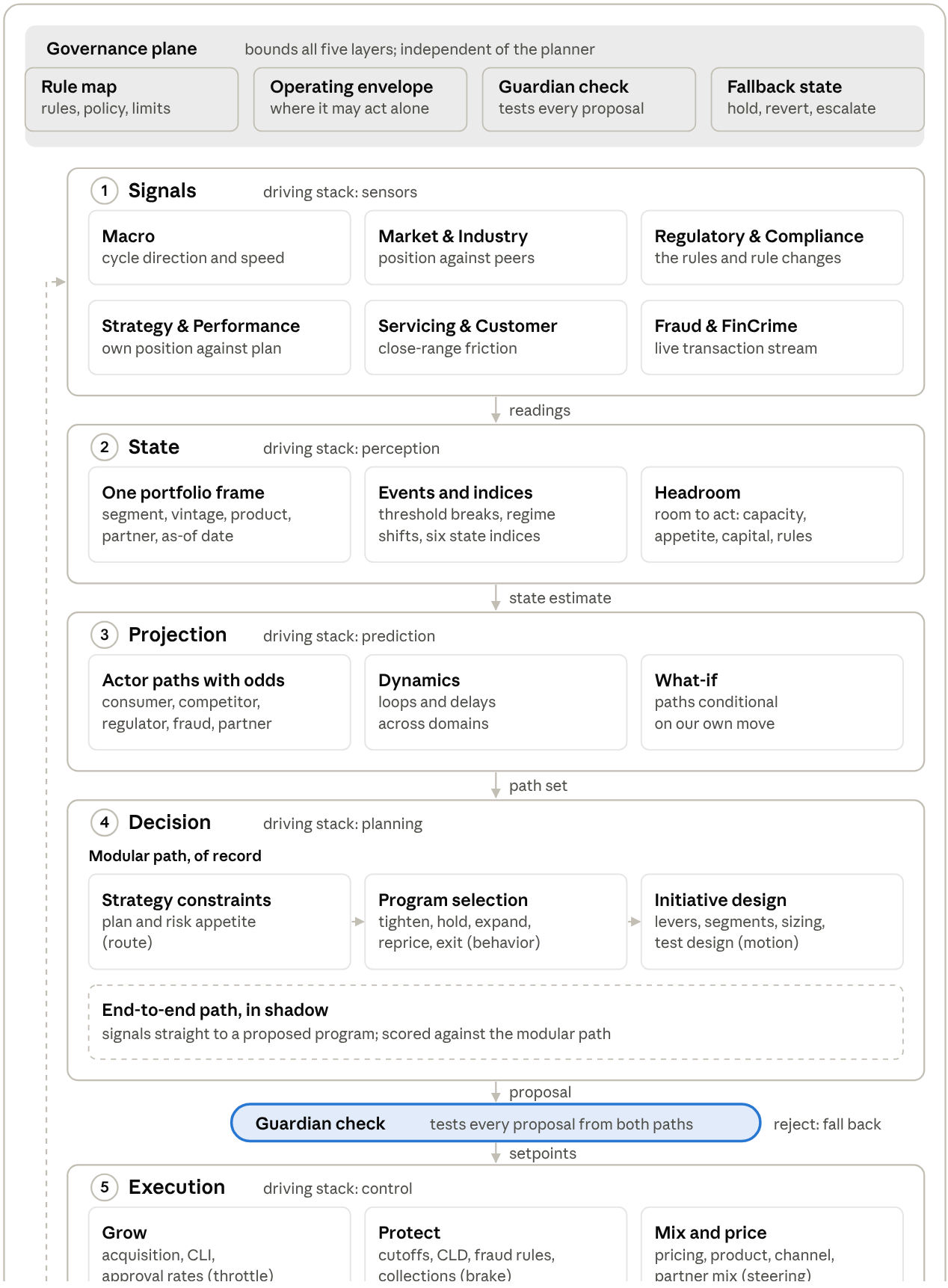
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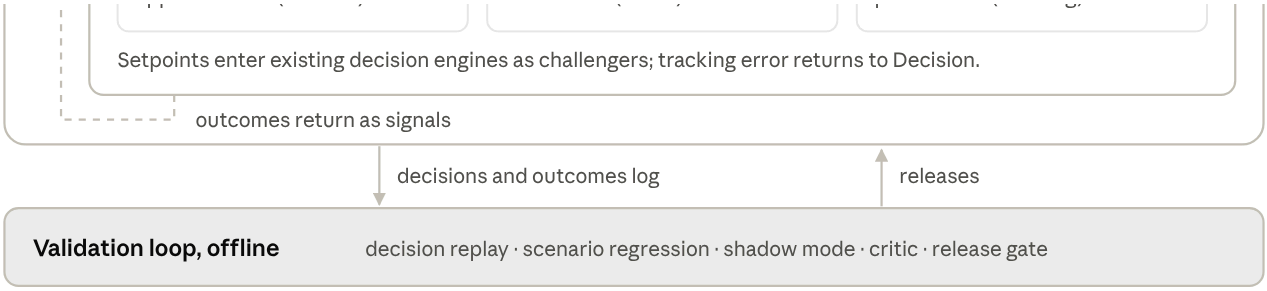
The driving stack maps onto signal intelligence layer for layer: sensors, perception, prediction, planning and control become signals, state, projection, decision and execution. Four changes follow for the CardsAI spine: synthesis splits into state and projection, execution becomes its own closed loop, governance becomes three named mechanisms, and validation becomes a standing loop.

The proposed stack

Same five layers as a driving stack, inside one governance plane

Signal intelligence stack for CardsAI; driving-stack equivalent in grey beside each layer





signal intelligence stack · 5 layers, governance plane, validation loop

Read top to bottom: each layer hands a typed object to the next, every proposal passes the guardian check before execution, and outcomes return as signals.

Layer mapping

Each driving layer has a direct equivalent, and each hands a typed object to the next. The split of synthesis into state and projection is not specific to driving: situation-awareness research separates comprehension of the current situation from projection of its future status, and both from the decision (Endsley and Rodgers, FAA 1994).

Driving stack	Signal intelligence	Hands down	Current spine → change
Sensors: cameras, radar, lidar, ultrasonic, GPS/IMU	**Signals**: 65 feeds across six domains	Readings with as-of time, freshness and lineage	Sense → add lineage and failure mode to the feed schema
Perception: objects, free space, own position	**State**: one fused picture of now: events, six state indices, headroom, position against plan	State estimate with confidence	Synthesize → split. The cockpit already separates State; add headroom
Prediction: paths with probabilities for each road user	**Projection**: paths with odds for each actor (consumer, competitor, regulator, fraud actor, partner), conditional on our own move	Path set per actor	Synthesize → split. Cockpit Dynamics and Scenario sit here, validated against outcomes separately from State

Driving stack	Signal intelligence	Hands down	Current spine → change
Planning: route, behavior, motion; modular or end-to-end	**Decision**: strategy constraints, program selection, initiative design, with an end-to-end path beside the modular one	Proposal: levers, segments, sizing, test design, expected response	Generate → two paths, modular of record
Control: steering, throttle, brake, at a higher rate than planning	**Execution**: setpoints into existing decision engines, tracked against expected response	Setpoints out; tracking error and test-versus-control readout back	Stage/Task → name a closed-loop layer
Safety envelope and fallback, independent of the planner	**Governance plane**: rule map, operating envelope, guardian check, fallback state	Pass, reject or fall back, with reasons	Govern → three mechanisms working from one rule map
Simulation, shadow mode, critic, offline	**Validation loop**: decision replay, scenario regression, shadow mode, critic	Release decision; override record	Experiment loop and Programs library → one standing loop

Headroom is the free-space equivalent. Perception reports drivable area as well as obstacles; State should report room to act (capacity, risk appetite, capital, regulatory) as well as threats.

Spine and cockpit terms are as last documented in this project. Where execution tracking already exists under Stage/Task, the change is naming it as a layer.

Signals as sensors

No domain is reliable alone. Each reads one thing well, goes blind in a known way, and needs a cross-check from an independent channel.

Domain	Closest sensor	Reads	Blind spot	Cross-check
Macro	Radar	Direction and speed of the cycle: labor, rates, credit conditions	National, monthly, revised, and able to go dark: October 2025 CPI survey data and household-survey data were never collected (BLS)	Own payment rates and roll rates by segment; private labor and spend data
Market & Industry	Lidar	Position against peers: pricing, offers, share	Sampled and lagged to filings; the precise feeds are paid	Own response rates, win/loss and attrition by competing offer
Regulatory & Compliance	Map and road signs	The rule map and changes to it: rulemaking, enforcement, exam focus	Meaning unclear until enforced; timing uncertain	Own complaint mix and audit findings against public enforcement themes
Strategy & Performance	GPS/IMU	Own position against plan: growth, margin, DQ, NCO	Nothing about surroundings; NCO confirms what broke two quarters earlier, since open-end credit charges off at 180 days past due (OCC Bulletin 2000-20)	Industry DQ and NCO; network volumes
Servicing & Customer	Ultrasonic	Close-range friction: calls, complaints, disputes	Only customers who make contact	CFPB complaint database; app-store and social

Domain	Closest sensor	Reads	Blind spot	Cross-check
Fraud & FinCrime	Cameras	Dense, live stream of transactions and attempts	Adversary adapts; confirmed-fraud labels arrive late	Network and consortium intelligence; dispute reason codes

The regulatory domain behaves less like a sensor than like the map: a maintained prior, with rulemaking and enforcement arriving as changes to it. Waymo makes the same case for maps as a prior that frees the system to attend to what is new (10 AI Lessons, Aug 2026).

The camera-only versus lidar argument has a direct equivalent: whether internal plus free public data is enough, or paid feeds are needed as the precise anchor. In the feed catalog as built, 27 of 36 external feeds are free or freemium; the paid tier buys speed and granularity in competitor and threat intelligence.

Design rules carried over

Six rules the driving industry converged on, each with its consequence for CardsAI.

1. Corroborate across independent channels, on one frame

Waymo fuses camera, lidar and radar over time into one set of objects and semantics before anything reasons on it (Demonstrably Safe AI, Dec 2025). Mobileye goes further and engineers two sensing systems independently, on the argument that inputs fused early behave as one system and fail as one (Shashua, 2018).

For CardsAI: project every signal onto one portfolio frame (segment, vintage, product, partner, as-of date) before combining. Count corroboration only across independent origins; two feeds built on the same bureau file or the same release are one reading. The internal and external mirror pairs already in the ontology are the redundant pairs.

2. Keep the proposer and the approver separate

Mobileye splits the policy that proposes an action from the formal rule set that approves or rejects it (Shashua, 2018). Waymo runs a separate validation layer over every path its generative model produces (Dec 2025). NVIDIA runs a classical safety stack in parallel

with its end-to-end stack (Kani, Jan 2026).

For CardsAI: this is "governance is a plane, not a peer" as a mechanism. Every proposal, from either path, passes a guardian check against the rule map: regulation, policy, risk appetite limits, partner contracts. The check is deterministic and owned outside the team that builds the planner. That is what effective challenge asks for: "critical analysis conducted by objective experts" (SR 26-2).

3. Run two paths and keep the intermediate objects

The modular-versus-end-to-end argument has not produced a winner. Tesla replaced its rule-based stack with a single network in 2024 (Nebot and Berrio Perez, 2026). Waymo consolidated into fewer, larger models but stopped short of a single black box (10 AI Lessons), keeping structured intermediate representations that can be checked (Dec 2025). The research middle path keeps the modules and trains them all toward the plan (Hu et al., CVPR 2023).

For CardsAI: the modular path is the path of record, with state, projection, options and decision stored as objects. An end-to-end path runs beside it, from signals straight to a proposed program, and is scored against it. Where the two disagree is the most useful validation data the system produces.

The cards-side version of the end-to-end path already exists at the networks. Visa Research's TREASURE and Mastercard's foundation model are single models trained on billions of transactions to serve many tasks (Yeh et al., 2025; TechInformed, 2026).

Regulatory position: SR 26-2 replaced SR 11-7 on April 17, 2026. It places generative and agentic AI outside its scope and leaves their controls to the bank's own risk governance (Federal Reserve). Stored intermediates, an independent check and shadow evidence are how the end-to-end path gets governed without a supervisory template.

4. Plan slow, control fast

Waymo pairs a fast sensor-fusion model for reactions with a slower vision-language model for unusual scenes, because the slow model cannot sit in the control loop (10 AI Lessons). Control then tracks the planned path at a higher rate than the planner replans.

For CardsAI: three loops, not one.

Loop	Cycle	Runs in	Signal intelligence role
Transaction	Milliseconds. Mastercard's Decision Intelligence Pro refines its risk score in under 50 ms (Mastercard, 2024)	Existing decision engines	Sets parameters; stays out of the path
Execution program	Daily signals, 5 to 14 day windows	Program tier	Owns it: setpoints, tracking, correction
Initiative	Monthly to quarterly	Initiative tier	Owns it: state, projection, proposal

Setpoints are cutoffs, thresholds, budgets and treatment tables. They enter the engines as challengers on a random slice, which is also what makes their effect measurable (Fahner, FICO). The levers group as the three actuators: grow (acquisition, CLI, approval rates), protect (cutoffs, CLD, fraud rules, collections), and mix and price.

5. Set the envelope and the fallback before granting autonomy

SAE J3016 ties each automated feature to the operating domain it was designed for. It requires a fallback to a minimal-risk condition on system failure or on leaving that domain. The line between levels is who performs the fallback: the human at Level 3, the system at Levels 4 and 5 (Brettin et al., quoting J3016).

Handing back to a disengaged human is the weak point. Drivers physically take over in under five seconds, but stay vulnerable to missing a critical event for a further 15 to 25 seconds (Schwarz et al., NHTSA ESV). The decision-system equivalent is an approver who has stopped reading.

Waymo's own conclusion is that improving driver assistance is a "false summit" on the way to driverless operation (10 AI Lessons). Supervised operation does not accumulate the evidence that unsupervised operation needs.

For CardsAI: give each decision class a level, an envelope and a fallback agreed in advance. Default assignment:

Decision class	Level	System may act when	Fallback
Strategy, one-way doors: pricing architecture, partner renewal, product exit	Advise	Never; it informs the decision	Leadership decides
Initiative: underwriting posture, CLI program sizing, marketing reallocation	Recommend; a person approves each	Never alone; a proposal reaches the approver once the guardian check passes	Hold current strategy
Execution, reversible: fraud threshold retune within band, collections capacity shift, channel budget shift within plan	Act inside the envelope; review after	Signals fresh and corroborated, regime inside tested range, exposure under cap, move reversible inside the window	Revert to champion, freeze changes, alert the owner
Transaction: approve or decline, line assignment	Unchanged	Signal intelligence sets parameters, never account decisions	Existing engine controls

The initiative row is the one to watch. A person approving every recommendation is supervised automation, and attention degrades as trust builds. Either the approver gets the full state behind each proposal, or the class moves to a narrow envelope the system owns.

6. Treat validation as a standing loop

Waymo builds a Driver, a Simulator and a Critic from one foundation model (Dec 2025). The Critic exists so the Driver does not mark its own work (10 AI Lessons). Tesla has run its software silently beside human drivers since 2016, logging where the two disagree (IEEE Spectrum).

Driving practice	Signal intelligence equivalent	Measure
Log replay	Decision replay: past decision points re-run on point-in-time data	Was the proposal right on what was known then
Scenario library	The cockpit scenarios as a regression suite: credit turn in growth vintages, partner portfolio drift, premium squeeze, fraud-friction loop	Pass or fail per scenario, per release
Shadow mode	System proposes silently beside the current decision process	Divergence rate; who was right once outcomes land
Critic	Independent scoring of decisions, the system's and people's, against outcomes	Override rate and override quality
Fleet log	Programs library: each decision with its signals, projection, expected and actual response	Share of decision classes covered

SR 26-2 asks for the same instrument on models in scope: outcomes analysis, comparing outputs with real-world results (Federal Reserve).

Where the comparison breaks

The stack carries over; the physics does not. Eight differences change the design.

Driving	Cards	Design consequence
Feedback arrives in milliseconds	Credit outcomes take quarters: charge-off comes 180 days after the missed payment	Steer on leading indicators (early DQ, roll rates, payment rate) and holdouts; the loop closes slowly and only in part

Driving	Cards	Design consequence
Vehicle physics is stable and known	Response to a lever is statistical, and customers, competitors and fraud actors adapt to the move	Test and control is the measuring instrument; projections are ranges; closed-loop simulation has low fidelity
Perception has ground truth: an object is there or it is not	State is latent: consumer stress is never observed directly	State carries confidence and is validated through projections and outcomes
Experience is abundant: more than 200 million driverless miles at Waymo, tens of billions simulated (10 AI Lessons)	A handful of credit cycles	Rare regimes come from stress scenarios and industry history, not own data
One objective dominates: avoid harm	Growth, loss, customer experience and compliance trade off, and people set the weights	Risk appetite sets the cost function; the system does not learn it
Three actuators, one controller	Dozens of levers across functions, some with notice periods or no reverse	The envelope classifies levers by reversibility and lead time
A sensor reading stands once taken	Macro data is revised, and can be missing for good	Store point-in-time; replay uses what was known then
A learned policy can drive behind an independent checker	Adverse action on an account needs specific principal reasons (Reg B § 1002.9)	The end-to-end path proposes parameters and programs; governed models make account decisions

Sources

Pages opened for this comparison. CardsAI references (signal ontology, feed catalog, cockpit specification, business cases) are as built in this project.

Driving stacks

- Waymo, "Demonstrably Safe AI for Autonomous Driving", Dec 2025

- Waymo, "10 AI Lessons from Driving 200+ Million Fully Autonomous Miles", Aug 26, 2026
- Shashua, Mobileye opinion piece on true redundancy and Responsibility-Sensitive Safety, May 16, 2018
- Kani, NVIDIA blog on the DRIVE AV dual-stack architecture, Jan 22, 2026
- Nebot and Berrio Perez, "The Era of End-to-End Autonomy", arXiv preprint, Mar 2026
- Harris, "Tesla's Autopilot Depends on a Deluge of Data", IEEE Spectrum, Aug 2022
- Hu et al., "Planning-Oriented Autonomous Driving", CVPR 2023

Standards and human factors

- Brettin et al., "Teleoperation Operational Design Domain based on Minimal Risk Maneuver Capability", arXiv, quoting SAE J3016 (2021)
- Schwarz et al., "Transfer from Highly Automated to Manual Control: Performance and Trust", NHTSA ESV paper 17-0426
- Endsley and Rodgers, "Situation Awareness Information Requirements for En Route Air Traffic Control", FAA, Dec 1994

Cards and regulation

- Federal Reserve, SR 26-2, "Revised Guidance on Model Risk Management", Apr 17, 2026
- CFPB, Regulation B § 1002.9, Notifications
- OCC Bulletin 2000-20, adopting the FFIEC Uniform Retail Credit Classification and Account Management Policy, Jun 2000
- BLS, revised news release dates following the 2025 lapse in appropriations
- Mastercard, "Mastercard supercharges consumer protection with gen AI", Feb 1, 2024
- Yeh et al., Visa Research, "TREASURE: A Transformer-Based Foundation Model for High-Volume Transaction Understanding", arXiv, Nov 2025
- TechInformed on Mastercard's payments foundation model, Mar 2026
- Fahner, FICO, "Causal Modeling-Based Approach for Testing and Improving Credit Decisions Over Time", Aug 2011